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EIGENVALUES OF DUAL HERMITIAN MATRICES WITH APPLICATION IN FORMATION CONTROL*

LIQUN QI† AND CHUNFENG CUI‡

Abstract. We propose a supplement matrix method for computing eigenvalues of a dual Hermitian matrix and discuss its application in multiagent formation control. Suppose we have a ring, which can be the real field, the complex field, or the quaternion ring. We study dual number symmetric matrices, dual complex Hermitian matrices, and dual quaternion Hermitian matrices in a unified frame of dual Hermitian matrices. An $n \times n$ dual Hermitian matrix has n dual number eigenvalues. We define determinant, characteristic polynomial, and supplement matrices for a dual Hermitian matrix. Supplement matrices are Hermitian matrices in the original ring. The standard parts of the eigenvalues of that dual Hermitian matrix are the eigenvalues of the standard part of the dual Hermitian matrix in the original ring, while the dual parts of the eigenvalues of that dual Hermitian matrix are the eigenvalues of those supplement matrices. Note that the standard part of the dual Hermitian matrix is still a Hermitian matrix in the original ring. Hence, by applying any practical method for computing eigenvalues of Hermitian matrices in the original ring, we have a practical method for computing eigenvalues of a dual Hermitian matrix. We call this method the supplement matrix method. In multiagent formation control, a desired relative configuration scheme may be given. People need to know if this scheme is reasonable such that a feasible solution of configurations of these multiagents exists. By exploring the eigenvalue problem of dual Hermitian matrices and its link with the unit gain graph theory, we open a cross-disciplinary approach to solve the relative configuration problem. Numerical experiments are reported to show the efficiency of our proposed methods.

Key words. dual number, dual quaternion, dual Hermitian matrix, eigenvalue, supplement matrix, formation control

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1. Introduction.

Dual numbers, dual quaternions, their vectors and matrices, as well as their applications have a long history. It was British mathematician William Kingdon Clifford who introduced dual numbers in 1873 [10]. Then German mathematician Eduard Study introduced dual angles in 1903 [41]. These started the study and applications of dual numbers, dual number vectors, and dual number matrices in kinematics, dynamics, and robotics [1, 17, 19, 31, 32, 42, 43]. Later, dual quaternions, in particular, unit dual quaternions, found wide applications in hand-eye calibration, neuroscience, multiagent formation control, and simultaneous location and mapping (SLAM), etc., [4, 5, 6, 13, 20, 26, 35, 38, 44]. Among these, dual quaternion matrices are used in multiagent formation control [35, 38]. Very recently, dual complex matrices found applications in brain science [45].

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†Department of Mathematics, School of Science, Hangzhou Dianzi University, Hangzhou 310018, China, and Department of Applied Mathematics, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong (maqilq@polyu.edu.hk).
‡Corresponding author. LMIB of the Ministry of Education, School of Mathematical Sciences, Beihang University, Beijing 100191, China (chunfengcui@buaa.edu.cn).

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For real and complex matrices, the eigenvalue problem is one major problem area in matrix computations [18], which is important in determining the stability of a time dependent differential equation and analyzing numerical methods [9]. The studies of the eigenvalue problems, including the generalized eigenvalue problem, the perturbation theory, and the inverse eigenvalue problem [2, 7, 8, 18], have achieved great progress. However, the study of the eigenvalue problem for dual matrices is still in its early stage and requires further exploration and analysis. In 2007, Pennestrì and Stefanelli [31] proposed the problem of computing eigenvalues and singular values of dual number matrices. Also see [32]. In 2023, Qi and Luo [37] showed that an $n \times n$ dual quaternion Hermitian matrix has n dual number eigenvalues and established the singular value decomposition of a dual quaternion matrix. These also apply to dual number matrices and dual complex matrices [34]. The relative configuration adjacent matrices and Laplacian matrices in multiagent formation control are dual quaternion Hermitian matrices [35, 38].

Several numerical methods for computing eigenvalues of dual quaternion Hermitian matrices arose. These include a power method [12], a bidiagonalization method [14], and a Rayleigh quotient iteration method [15].

We study dual number symmetric matrices, dual complex Hermitian matrices, and dual quaternion Hermitian matrices in a unified frame of dual Hermitian matrices. Suppose we have a ring, which can be the real field, the complex field, or the quaternion ring. Then an $n \times n$ dual Hermitian matrix has n dual number eigenvalues. The trouble of finding these n dual number eigenvalues occurs when the standard part of the dual Hermitian matrix has multiple eigenvalues. In this case, the characteristic polynomial of that dual Hermitian matrix has infinitely many roots, which are not eigenvalues of that dual Hermitian matrix in general. This may make a computational method divergent or slow.

In this paper, we define supplement matrices for a dual Hermitian matrix. Supplement matrices are Hermitian matrices in the original ring. The standard parts of the eigenvalues of that dual Hermitian matrix are the eigenvalues of the standard part Hermitian matrix in the original ring, while the dual parts of the eigenvalues of that dual Hermitian matrix are the eigenvalues of those supplement matrices. Hence, by apply any practical method for computing eigenvalues of Hermitian matrices in the original ring, we have a practical method for computing eigenvalues of a dual Hermitian matrix. We call this method the *supplement matrix method*.

Then we study the relative configuration problem in multiagent formation control. People need to know if a given desired relative configuration scheme is reasonable such that a feasible solution of configurations of these multiagents exists. By combining the eigenvalue problem of dual Hermitian matrices with the unit gain graph theory, we open a cross-disciplinary approach to solve the relative configuration problem. Then the supplement matrix method is used for this approach.

In the next section, we review some basic knowledge of dual elements and dual matrices in a ring. That ring can be the real field, the complex field, or the quaternion ring. We also review some knowledge about unit dual quaternions there. This is useful for studying multiagent formation control. In section 3, we define the determinants and characteristic polynomials of dual Hermitian matrices and show that when the standard part of a dual Hermitian matrix has multiple eigenvalues, the characteristic polynomial of that dual Hermitian matrix may have infinitely many roots, which are not eigenvalues of that dual Hermitian matrix in general. We define supplement matrices, construct the supplement matrix method, and prove that it does find all eigenvalues of a dual Hermitian matrix in section 4. In section 5, we study the

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relative configuration problem in formation control. We explore the cross-disciplinary approach to solve the relative configuration problem in section 6. Numerical experiments are reported in section 7. It is shown that when the dimension of the problem is big, the supplement matrix method is faster and more accurate than the power method. Some concluding remarks are made in section 8.

2. Dual elements and dual matrices.

2.1. Dual elements and unit dual quaternions. The field of real numbers, the field of complex numbers, and the ring of quaternions are denoted by $\mathbb{R}, \mathbb{C}$, and $\mathbb{Q}$, respectively. We use $\mathbb{G}$ to represent them. Thus, $\mathbb{G}$ may be $\mathbb{R}$ or $\mathbb{C}$ or $\mathbb{Q}$.

We use $\hat{\mathbb{R}}, \hat{\mathbb{C}}$, and $\hat{\mathbb{Q}}$ to denote the ring of dual numbers, dual complex numbers, and dual quaternions, respectively, and use $\hat{\mathbb{G}}$ to represent them in general. We call an element in $\hat{\mathbb{G}}$ a dual element**.** Thus a dual element means a dual number, or a dual complex number, or a dual quaternion, depending upon $\hat{\mathbb{G}} = \hat{\mathbb{R}}$, or $\hat{\mathbb{C}}$, or $\hat{\mathbb{Q}}$. Similarly, a Hermitian matrix in $\mathbb{G}$ means a symmetric matrix, or a complex Hermitian matrix, or a quaternion Hermitian matrix, depending upon $\mathbb{G} = \mathbb{R}$, or $\mathbb{C}$, or $\mathbb{Q}$.

Our arguments can also be generalized to the other rings. We do not pursue this here.

A dual element $a = a_s + a_d\epsilon \in \hat{\mathbb{G}}$ has a standard part $a_s \in \mathbb{G}$ and a dual part $a_d \in \mathbb{G}$. The symbol ϵ is the infinitesimal unit, satisfying $\epsilon^2 = 0$, and ϵ is commutative with numbers in $\mathbb{G}$. The *conjugate* of a is defined as $a^* = a_s^* + a_d^*\epsilon$, where a_s^* and a_d^* is the conjugates of numbers a_s and a_d, respectively. Note that if a_s and a_d are real numbers, then their conjugates are themselves. Thus, the conjugate of a dual number is also itself. If $a_s \neq 0$, then we say that a is *appreciable*. Otherwise, we say that a is *infinitesimal*.

Suppose we have two dual elements $a = a_s + a_d\epsilon$ and $b = b_s + b_d\epsilon$. Then their sum is $a + b = (a_s + b_s) + (a_d + b_d)\epsilon$, and their product is $ab = a_s b_s + (a_s b_d + a_d b_s)\epsilon$. In this way, $\hat{\mathbb{G}}$ is a ring. In particular, $\hat{\mathbb{R}}$ and $\hat{\mathbb{C}}$ are two commutative rings, while $\hat{\mathbb{Q}}$ is a noncommutative ring.

If both a_s and a_d are real numbers, then $a = a_s + a_d\epsilon$ is called a dual number. Suppose we have two dual numbers $a = a_s + a_d\epsilon$ and $b = b_s + b_d\epsilon$. By [36], if $a_s > b_s$, or $a_s = b_s$ and $a_d > b_d$, then we say $a > b$. Then this defines positive, nonnegative dual numbers, etc.

For a dual element $a = a_s + a_d\epsilon \in \hat{\mathbb{G}}$, its magnitude is defined as a nonnegative dual number

$$|a| := \begin{cases} |a_s| + \dfrac{(a_s a_d^* + a_d a_s^*)}{2|a_s|}\epsilon & \text{if } a_s \neq 0, \\ |a_d|\epsilon & \text{otherwise.} \end{cases}$$

We use 0, $\mathbf{0}$, and O to denote a zero number, a zero vector, and a zero matrix, respectively.

A dual element $a = a_s + a_d\epsilon \in \hat{\mathbb{G}}$ is called invertible if there exists a dual element $b \in \hat{\mathbb{G}}$ such that $ab = ba = 1$. We can derive that a is invertible if and only if a is appreciable. In this case, we have

$$a^{-1} = a_s^{-1} - a_s^{-1} a_d a_s^{-1}\epsilon.$$

Let $q = q_s + q_d\epsilon \in \hat{\mathbb{Q}}$. If $|q| = 1$, then q is called a *unit dual quaternion*. A unit dual quaternion q is always invertible and we have $q^{-1} = q^*$. The three-dimensional motion of a rigid body can be represented by a unit dual quaternion. We have

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$$qq^* = (q_s + q_d\epsilon)(q_s^* + q_d^*\epsilon) = q_s q_s^* + (q_s q_d^* + q_d q_s^*)\epsilon = q^* q.$$

Thus, q is a unit dual quaternion if and only if q_s is a unit quaternion, and

$$q_s q_d^* + q_d q_s^* = q_s^* q_d + q_d^* q_s = 0. \tag{2.1}$$

Note that a quaternion is called an *imaginary quaternion* if its real part is zero. Suppose that there is a rotation $q_s \in \mathbb{Q}$ succeeded by a translation $p^b \in \mathbb{Q}$, where p^b is an imaginary quaternion. Here, following [44], we use the superscript b to represent the relation of the rigid body motion with respect to the body frame attached to the rigid body. Then the whole transformation can be represented using unit dual quaternion $q = q_s + q_d\epsilon$, where $q_d = \frac{1}{2} q_s p^b$. Note that we have

$$q_s q_d^* + q_d q_s^* = \frac{1}{2}\left[q_s (p^b)^* q_s^* + q_s p^b q_s^*\right] = \frac{1}{2} q_s \left[(p^b)^* + p^b\right] q_s^* = 0.$$

Thus, a transformation of a rigid body can be represented by a unit dual quaternion

$$q = q_s + \frac{\epsilon}{2} q_s p^b, \tag{2.2}$$

where q_s is a unit quaternion to represent the rotation, and p^b is an imaginary quaternion to represent the translation or the position. On the other hand, every attitude of a rigid body which is free to rotate relative to a fixed frame can be identified by a unique unit quaternion q. Thus, in (2.2), q_s is the attitude of the rigid body, while q_d represents the transformation. A unit dual quaternion q serves as both a specification of the configuration of a rigid body and a transformation taking the coordinates of a point from one frame to another via rotation and translation. In (2.2), if q is the configuration of the rigid body, then q_s and p^b are the attitude of and position of the rigid body, respectively. Denote the set of unit dual quaternions by $\hat{\mathbb{U}}$.

A *dual vector* is denoted by $\mathbf{x} = (x_1, \ldots, x_n)^\top \in \hat{\mathbb{G}}^n$. Its conjugate is $\mathbf{x}^* = (x_1^*, \ldots, x_n^*)$. We may denote $\mathbf{x} = \mathbf{x}_s + \mathbf{x}_d\epsilon$, where $\mathbf{x}_s, \mathbf{x}_d \in \mathbb{G}^n$. Its 2-norm is defined as

$$\|\mathbf{x}\|_2 = \begin{cases} \|\mathbf{x}_s\|_2 + \frac{\mathbf{x}_s^*\mathbf{x}_d + \mathbf{x}_d^*\mathbf{x}_s}{2\|\mathbf{x}_s\|_2}\epsilon & \text{if } \mathbf{x}_s \neq \mathbf{0}, \\ \|\mathbf{x}_d\|_2\epsilon & \text{if } \mathbf{x}_s = \mathbf{0}. \end{cases} \tag{2.3}$$

This is a dual number. For convenience in the numerical experiments, we also denote the 2^R-norm of a dual vector $\mathbf{x} = (x_i) \in \hat{\mathbb{G}}^n$ as

$$\|\mathbf{x}\|_{2^R} = \sqrt{\|\mathbf{x}_s\|_2^2 + \|\mathbf{x}_d\|_2^2},$$

which is a real number.

We say $\mathbf{x} = \mathbf{x}_s + \mathbf{x}_d\epsilon = (x_1, \ldots, x_n)^\top \in \hat{\mathbb{G}}^n$ is a *unit dual vector* if $\|\mathbf{x}\|_2 = 1$, or equivalently, $\|\mathbf{x}_s\|_2 = 1$ and $\mathbf{x}_s^*\mathbf{x}_d + \mathbf{x}_d^*\mathbf{x}_s = 0$. If $\mathbf{x}_s \neq \mathbf{0}$, then we say that $\mathbf{x}$ is appreciable. The unit vectors in $\mathbb{R}^n$ are denoted as $\mathbf{e}_1, \ldots, \mathbf{e}_n$. They are also unit vectors of $\mathbb{G}^n$ and $\hat{\mathbb{G}}^n$.

Let $\mathbf{x}, \mathbf{y} \in \hat{\mathbb{G}}^n$. If $\mathbf{x}^*\mathbf{y} = 0$, then we say that $\mathbf{x}$ and $\mathbf{y}$ are orthogonal. If $\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)} \in \hat{\mathbb{G}}^n$ and $\left(\mathbf{x}^{(i)}\right)^* \mathbf{x}^{(j)} = \delta_{ij}$ for $i, j = 1, \ldots, n$, where δ_{ij} is the Kronecker symbol, then we say that $\{\mathbf{x}^{(1)}, \cdots, \mathbf{x}^{(n)}\}$ is an orthonormal basis of $\hat{\mathbb{G}}^n$. Let $\mathbf{x}^{(1)} = \mathbf{x}_s^{(1)} + \mathbf{x}_d^{(1)}\epsilon, \ldots, \mathbf{x}^{(k)} = \mathbf{x}_s^{(k)} + \mathbf{x}_d^{(k)}\epsilon \in \hat{\mathbb{G}}^n$. If $\mathbf{x}_s^{(1)}, \ldots, \mathbf{x}_s^{(k)}$ are linearly independent, then we say that $\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(k)}$ are appreciably linearly independent.

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2.2. Dual matrices. Assume that $A = A_s + A_d\epsilon$ and $B = B_s + B_d\epsilon$ are two dual matrices in $\hat{\mathbb{G}}^{n\times n}$, where n is a positive integer, $A_s, A_d, B_s, B_d \in \mathbb{G}^{n\times n}$. If $AB = BA = I$, where I is the $n\times n$ identity matrix, then we say that B is the *inverse* of A and denote that $B = A^{-1}$.

For a dual matrix $A \in \hat{\mathbb{G}}^{n\times n}$, denote its conjugate transpose as A^*. If $A^* = A$, then A is called a *dual Hermitian matrix*. If $A^* = A^{-1}$, then A is called a *dual unitary matrix*. In particular, if A is a dual Hermitian matrix in $\hat{\mathbb{R}}^{n\times n}$, or $\hat{\mathbb{C}}^{n\times n}$, or $\hat{\mathbb{Q}}^{n\times n}$, respectively, then A is called a dual number symmetric matrix, or a dual complex Hermitian matrix, or a dual quaternion Hermitian matrix, respectively. If A is a dual matrix in $\hat{\mathbb{R}}^{n\times n}$, or $\hat{\mathbb{C}}^{n\times n}$, or $\hat{\mathbb{Q}}^{n\times n}$, respectively, then A is called a dual number matrix, or a dual complex matrix, or a dual quaternion matrix, respectively.

Let $k \le n$. We say that $A \in \hat{\mathbb{G}}^{n\times k}$ is *partially unitary* if its column vectors are unit dual vectors and orthogonal to each other.

The F-norm of a dual matrix $A = (a_{ij}) \in \hat{\mathbb{G}}^{m\times n}$ is

$$\|A\|_F = \begin{cases} \|A_s\|_F + \frac{tr(A_s^*A_d + A_d^*A_s)}{2\|A_s\|_F}\epsilon & \text{if } A_s \neq O, \\ \|A_d\|_F\epsilon & \text{otherwise.} \end{cases}$$

This is a dual number. For convenience in the numerical experiments, we also define the F^R-norm of a dual matrix $A = (a_{ij}) \in \hat{\mathbb{G}}^{m\times n}$ as

$$\|A\|_{F^R} = \sqrt{\|A_s\|_F^2 + \|A_d\|_F^2},$$

which is a real number.

It is classical that an $n\times n$ real symmetric or complex Hermitian matrix has n real eigenvalues, and this matrix is positive semidefinite (or definite, respectively) if and only if all of these n eigenvalues are nonnegative (or positive, respectively). In 1997, Zhang [47] extended this to quaternion Hermitian matrices. In 2023, Qi and Luo [37] further extended this to dual quaternion Hermitian matrices. This is actually also true for dual symmetric or dual complex Hermitian matrices. We now state these in our general frame.

Let $A \in \hat{\mathbb{G}}^{n\times n}$ be a dual Hermitian matrix and $\mathbf{x} \in \hat{\mathbb{G}}^n$. Then $\mathbf{x}^*A\mathbf{x}$ is a dual number if $\hat{\mathbb{G}}$ is either $\hat{\mathbb{R}}$, or $\hat{\mathbb{C}}$, or $\hat{\mathbb{Q}}$. Thus, by [36], we may distinguish that $\mathbf{x}^*A\mathbf{x}$ is nonnegative, or positive, or not. If for all $\mathbf{x} \in \hat{\mathbb{G}}^n$, $\mathbf{x}^*A\mathbf{x}$ is nonnegative, then we say that A is positive semidefinite. If for all $\mathbf{x} \in \hat{\mathbb{G}}^n$ and appreciable, $\mathbf{x}^*A\mathbf{x}$ is positive, then we say that A is positive definite.

Let $A \in \hat{\mathbb{G}}^{n\times n}$, $\mathbf{x} \in \hat{\mathbb{G}}^n$ be appreciable, and $\lambda \in \hat{\mathbb{G}}$. If

$$A\mathbf{x} = \mathbf{x}\lambda, \tag{2.4}$$

then λ is called a right eigenvalue of A, with $\mathbf{x}$ as its corresponding right eigenvector. If

$$A\mathbf{x} = \lambda\mathbf{x}, \tag{2.5}$$

where $\mathbf{x}$ is appreciable, i.e., $\mathbf{x}_s \neq \mathbf{0}$, then λ is called a left eigenvalue of A, with a left eigenvector $\mathbf{x}$.

If $\hat{\mathbb{G}}$ is $\hat{\mathbb{R}}$ or $\hat{\mathbb{C}}$, then the multiplication is commutative. In these two cases, it is not necessary to distinguish right and left eigenvalues. We just call them eigenvalues [34]. It was proved in [37] that all the right eigenvalues of a dual quaternion Hermitian

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matrix A are dual numbers. As dual numbers are commutative with dual quaternions, they are also left eigenvalues. Thus, we may simply call them *eigenvalues* of A. Note that A may still have other left eigenvalues, which are not dual numbers. See an example of a quaternion matrix in [47].

The following theorem was proved in [37] for dual quaternion Hermitian matrices. It is also true for dual symmetric matrices and dual complex Hermitian matrices by similar arguments [34].

THEOREM 2.1. *Suppose that $A \in \hat{\mathbb{G}}^{n\times n}$ is a dual symmetric matrix, or a dual complex Hermitian matrix, or a dual quaternion Hermitian matrix, and then it has exactly n dual number eigenvalues. It is positive semidefinite or definite if and only if these n eigenvalues are nonnegative or positive.*

Write $A = A_s + A_d\epsilon$, $\lambda = \lambda_s + \lambda_d\epsilon$, and $\mathbf{x} = \mathbf{x}_s + \mathbf{x}_d\epsilon$. Then (2.5) is equivalent to

$$A_s\mathbf{x}_s = \lambda_s\mathbf{x}_s \tag{2.6}$$

with $\mathbf{x}_s \neq \mathbf{0}$, i.e., λ_s is an eigenvalue of A_s with an eigenvector $\mathbf{x}_s$, and

$$(A_s - \lambda_s I)\mathbf{x}_d = \lambda_d\mathbf{x}_s - A_d\mathbf{x}_s. \tag{2.7}$$

3. The characteristic polynomial of a dual Hermitian matrix. Suppose that $A \in \hat{\mathbb{G}}^{n\times n}$ is an $n \times n$ dual Hermitian matrix. Here, $\hat{\mathbb{G}}$ is either $\hat{\mathbb{R}}$, or $\hat{\mathbb{C}}$, or $\hat{\mathbb{Q}}$. Then A has n dual number eigenvalues $\lambda_1, \ldots, \lambda_n$. We may define its determinant as

$$\det(A) = \lambda_1\lambda_2 \ldots \lambda_n. \tag{3.1}$$

Furthermore, we may define the characteristic polynomial as

$$\phi(\lambda) = \det(\lambda I - A),$$

where I is the $n \times n$ identity matrix. This shows that ϕ is a dual number polynomial. Note that the factorization form of a dual number polynomial is not unique. For example, polynomial $\lambda^2 = (\lambda + a\epsilon)(\lambda - a\epsilon)$ for any real number a. This leads the characteristic polynomial of the dual Hermitian matrix to be more complicated than that of the Hermitian matrix.

THEOREM 3.1. *Suppose that $A = A_s + A_d\epsilon \in \hat{\mathbb{G}}^{n\times n}$ is an $n \times n$ dual Hermitian matrix. Here, $\hat{\mathbb{G}}$ is either $\hat{\mathbb{R}}$, or $\hat{\mathbb{C}}$, or $\hat{\mathbb{Q}}$. Let its n dual number eigenvalues be $\lambda_1, \ldots, \lambda_n$. Then its characteristic polynomial has the form*

$$\phi(\lambda) \equiv (\lambda - \lambda_1) \ldots (\lambda - \lambda_n). \tag{3.2}$$

Furthermore, a dual number $\lambda = \lambda_s + \lambda_d\epsilon$ is a root of ϕ, either if λ is an eigenvalue of A or if λ_s is a multiple eigenvalue of A_s.

Proof. First, we see that $\lambda I - A$ is an $n \times n$ dual Hermitian matrix too. Since A has n dual number eigenvalues $\lambda_1, \ldots, \lambda_n$, by the definition of eigenvalues of dual Hermitian matrices, we see that $\lambda I - A$ has n dual number eigenvalues $\lambda - \lambda_1, \ldots, \lambda - \lambda_n$. By the definition of determinants and characteristic polynomials, we have (3.2).

Clearly, any eigenvalue of A is a root of ϕ. By (2.6), the standard part of an eigenvalue of A is an eigenvalue of A_s. Let $\lambda = \lambda_s + \lambda_d\epsilon$ be a dual number. If λ_s is not an eigenvalue of A_s, then $\lambda - \lambda_i$ is an appreciable dual number for $i = 1, \ldots, n$.

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By (3.2), this implies $\phi(\lambda) \neq 0$, i.e., λ is not a root of ϕ. If λ_s is a single eigenvalue of A_s, then λ_s is equal to the standard part of λ_i for one i. Then $\lambda - \lambda_j$ is an appreciable dual number for $j \neq i$. By (3.2), $\phi(\lambda) = 0$ if and only if $\lambda = \lambda_i$. Finally, assume that λ_s is a multiple eigenvalue of A_s. Without loss of generality, we may assume that both the standard parts of λ_1 and λ_2 are λ_s. Let $\lambda_1 = \lambda_s + \lambda_{d1}\epsilon$, $\lambda_2 = \lambda_s + \lambda_{d2}\epsilon$, and $\lambda = \lambda_s + \lambda_d\epsilon$. Then

$$(\lambda - \lambda_1)(\lambda - \lambda_2) = (\lambda_d - \lambda_{d1})(\lambda_d - \lambda_{d2})\epsilon^2 = 0,$$

which implies that $\phi(\lambda) = 0$. This completes the proof. ∎

Theorem 3.1 reveals that when A_s has multiple eigenvalues, the characteristic polynomial of A may have infinitely many roots, which are not eigenvalues of A in general. Therefore, it will be difficult to find eigenvalues of A by the characteristic polynomial if we treat the standard parts and dual parts of the eigenvalues of A together. Another way is to find eigenvalues and eigenvectors of A_s by a classical method first. Then, the questions are, if λ_s is a single eigenvalue of A_s, can we give a formula of λ_d such that $\lambda_s + \lambda_d\epsilon$ is an eigenvalue of A? And if λ_s is a k-multiple eigenvalue of A_s, how can we find the dual parts of corresponding k eigenvalues of A? In the next section, we will answer these two questions.

In the following, we present a simple example to show that computing the standard part by (2.6) and the dual part by (2.7) separately may fail to get the eigenvalues and eigenvectors.

Example 3.2. Consider the dual matrix

$$A = \begin{bmatrix} 1 & \epsilon \\ \epsilon & 1 \end{bmatrix}.$$

For the standard part, there is $A_s = I$ and $\lambda_s = 1$.

Suppose $\mathbf{x}_{s1} = [1, 0]^\top$ and $\mathbf{x}_{s2} = [0, 1]^\top$. Then we have $A_s\mathbf{x}_{s1} = \lambda_s\mathbf{x}_{s1}$ and $A_s\mathbf{x}_{s2} = \lambda_s\mathbf{x}_{s2}$. Substituting $(\lambda_s, \mathbf{x}_{s1})$ and $(\lambda_s, \mathbf{x}_{s2})$ into (2.7), we have

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \lambda_{d1} \\ -1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ \lambda_{d2} \end{bmatrix},$$

respectively, which are unsolvable.

Suppose $\tilde{\mathbf{x}}_{s1} = \frac{\sqrt{2}}{2}[1, 1]^\top$ and $\tilde{\mathbf{x}}_{s2} = \frac{\sqrt{2}}{2}[1, -1]^\top$. Then both $(\lambda_s, \tilde{\mathbf{x}}_{s1})$ and $(\lambda_s, \tilde{\mathbf{x}}_{s2})$ satisfy (2.6). Substituting $(\lambda_s, \tilde{\mathbf{x}}_{s1})$ and $(\lambda_s, \tilde{\mathbf{x}}_{s2})$ into (2.7), we have

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \frac{\sqrt{2}}{2}\begin{bmatrix} \lambda_{d1} - 1 \\ -1 + \lambda_{d1} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \frac{\sqrt{2}}{2}\begin{bmatrix} \lambda_{d2} + 1 \\ -1 - \lambda_{d2} \end{bmatrix},$$

respectively. Hence, we obtain $\lambda_{d1} = 1$ and $\lambda_{d2} = -1$, and the dual parts of the eigenvectors $\mathbf{x}_{d1}$ and $\mathbf{x}_{d2}$ are arbitrary.

Example 3.2 shows that solving (2.6) and (2.7) separately may leave (2.7) unsolvable, even though the original system (2.5) is consistent. The reason is that when λ_s is a k-multiple eigenvalue of the standard part A_s, its corresponding eigenvectors span a k-dimensional linear space. Any vector in this space is an eigenvector of A_s corresponding to λ_s, but only some of them are the standard parts of the eigenvectors of A corresponding to an eigenvalue λ. In the above example, $\tilde{\mathbf{x}}_{s1}$ and $\tilde{\mathbf{x}}_{s2}$ are the standard parts of the eigenvectors of A, yet $\mathbf{x}_{s1}$ and $\mathbf{x}_{s2}$ are not. Now the question is how to choose eigenvectors of A_s in the k-dimensional linear space such that they are

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also the standard parts of the eigenvectors of A. Besides solving (2.6) and (2.7) separately, one may solve these two systems (2.6) and (2.7) together directly. However, it may be expensive since the overall system is nonlinear and may lose the special structure of the eigenvalue problem.

In the next section, we will exploit the eigenvalue problem of dual Hermitian matrices by categorizing the eigenvalues of A_s into the single eigenvalue case and the multiple eigenvalues case. We then present the formulations of the standard parts of the eigenvectors of A by the eigenvectors of A_s in the k-dimensional linear space.

4. A practical method for computing eigenvalues of a dual Hermitian matrix. In this section, we define supplement matrices for a dual Hermitian matrix $A \in \hat{\mathbb{G}}^{n\times n}$ to enable the calculation of the eigenvalues of A. Assume that $\mathbb{G}$ is either $\mathbb{R}$ or $\mathbb{C}$ or $\mathbb{Q}$.

THEOREM 4.1. *Suppose that $A = A_s + A_d\epsilon \in \hat{\mathbb{G}}^{n\times n}$ is a dual Hermitian matrix, and the real number λ_s is a k-multiple eigenvalue of the Hermitian matrix $A_s \in \mathbb{G}^{n\times n}$.*

If $k = 1$, i.e., λ_s is a single eigenvalue of A_s, let $\mathbf{x}_s$ be a unit eigenvector of A_s, associated with λ_s. Then $\lambda = \lambda_s + \lambda_d\epsilon$ is a single eigenvalue of A, where $\lambda_d = \mathbf{x}_s^* A_d \mathbf{x}_s$, with an eigenvector $\mathbf{x} = \mathbf{x}_s + \mathbf{x}_d\epsilon$, where $\mathbf{x}_d$ is a solution of

$$(A_s - \lambda_s I)\mathbf{x}_d = (\lambda_d I - A_d)\mathbf{x}_s. \tag{4.1}$$

If $k > 1$, let $\mathbf{v}_1, \ldots, \mathbf{v}_k$ be k orthonormal eigenvectors of A_s, associated with λ_s. Let $W = (\mathbf{v}_1 \ldots, \mathbf{v}_k)$. Then W is an $n \times k$ partially unitary matrix, and $W^* A_d W \in \mathbb{G}^{k\times k}$ is a Hermitian matrix. Let $\lambda_{d1}, \ldots, \lambda_{dk}$ be the k eigenvalues of $W^* A_d W$, with corresponding eigenvectors $\mathbf{y}_1, \ldots, \mathbf{y}_k$. Then $\lambda_i = \lambda_s + \lambda_{di}\epsilon$ for $i = 1, \ldots, k$ are eigenvalues of A, with eigenvectors $\mathbf{x}_i = \mathbf{x}_{si} + \mathbf{x}_{di}\epsilon$ for $i = 1, \ldots, k$, where $\mathbf{x}_{si} = W\mathbf{y}_i$, and $\mathbf{x}_{di}$ is a solution of

$$(A_s - \lambda_s I)\mathbf{x}_{di} = (\lambda_{di} I - A_d)\mathbf{x}_{si} \tag{4.2}$$

for $i = 1, \ldots, k$.

Proof. Suppose that $k = 1$, i.e., λ_s is a single eigenvalue of A_s. Let $\mathbf{x}_s$ be a unit eigenvector of A_s, associated with λ_s. Multiply (2.7) by $\mathbf{x}_s^*$ from the left. Then we have $\lambda_d = \mathbf{x}_s^* A_d \mathbf{x}_s$. From (2.7), we have (4.1).

Suppose that $k > 1$. Let $\mathbf{v}_1, \ldots, \mathbf{v}_k$ be k orthonormal eigenvectors of A_s, associated with λ_s. Let $W = (\mathbf{v}_1 \ldots, \mathbf{v}_k)$. Then W is an $n \times k$ partially unitary matrix, and $W^* A_d W \in \mathbb{G}^{k\times k}$ is a Hermitian matrix. Let $\lambda_{d1}, \ldots, \lambda_{dk}$ be the k eigenvalues of $W^* A_d W$, with corresponding eigenvectors $\mathbf{y}_1, \ldots, \mathbf{y}_k$. Then we have $W^* A_d W \mathbf{y}_i = \lambda_{di}\mathbf{y}_i$ for $i = 1, \ldots, k$. Let $\mathbf{x}_{si} = W\mathbf{y}_i$ for $i = 1, \ldots, k$. Then we have $A_s \mathbf{x}_{si} = \lambda_s \mathbf{x}_{si}$, i.e., $\mathbf{x}_{si}$ is an eigenvector of A_s, associated with eigenvalue λ_s for $i = 1, \ldots, k$. Now (2.7) has the form

$$(A_s - \lambda_s I)\mathbf{x}_{di} - \lambda_d \mathbf{x}_{si} = -A_d \mathbf{x}_{si}$$

for $i = 1, \ldots, k$. Multiply the above equality by $\mathbf{x}_{si}^*$ from the left. Then we have $\lambda_{di} = \mathbf{x}_{si}^* A_d \mathbf{x}_{si} = \mathbf{y}_i^* W^* A_d W \mathbf{y}_i$, and $\lambda_i = \lambda_s + \lambda_{di}\epsilon$ for $i = 1, \ldots, k$ are eigenvalues of A, with eigenvectors $\mathbf{x}_i = \mathbf{x}_{si} + \mathbf{x}_{di}\epsilon$, where $\mathbf{x}_{di}$ satisfies (4.2) for $i = 1, \ldots, k$.

This completes the proof. ∎

We call the Hermitian matrix $W^* A_d W \in \mathbb{G}^{k\times k}$ the *supplement matrix* of the dual Hermitian matrix A, corresponding to the k-multiple eigenvalue λ_s of A_s, the standard part of A.

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Then we have a practical method for computing eigenvalues of the dual Hermitian matrix A.

ALGORITHM 4.2. *The supplement matrix method. Suppose that* $A = A_s + A_d\epsilon \in \hat{\mathbb{G}}^{n\times n}$ *is a dual Hermitian matrix.*

Step 1. *Use a practical method to find* n *real eigenvalues of the Hermitian matrix* $A_s \in \mathbb{G}^{n\times n}$ *with a set of orthonormal eigenvectors.*

Step 2. *Assume that* λ_s *is a single eigenvalue of* A_s *with a unit vector* $\mathbf{x}_s$. *Then* $\lambda = \lambda_s + \lambda_d\epsilon$ *is a single eigenvalue of* A, *where* $\lambda_d = \mathbf{x}_s^* A_d \mathbf{x}_s$, *with an eigenvector* $\mathbf{x} = \mathbf{x}_s + \mathbf{x}_d\epsilon$, *where* $\mathbf{x}_d$ *is a solution of* (4.1).

Step 3. *Assume that* λ_s *is a* k*-multiple eigenvalue of* A_s *with* k *orthonormal eigenvectors* $\mathbf{v}_1, \ldots, \mathbf{v}_k$, *for* $k > 1$. *Let* $W = (\mathbf{v}_1 \ldots, \mathbf{v}_k)$ *and* $W^* A_d W \in \mathbb{G}^{k\times k}$. *Use a practical method to find* k *eigenvalues of* $W^* A_d W$, *as* $\lambda_{d1}, \ldots, \lambda_{dk}$, *with corresponding eigenvectors* $\mathbf{y}_1, \ldots, \mathbf{y}_k$. *Then* $\lambda_i = \lambda_s + \lambda_{di}\epsilon$ *for* $i = 1, \ldots, k$ *are eigenvalues of* A, *with eigenvectors* $\mathbf{x}_i = \mathbf{x}_{si} + \mathbf{x}_{di}\epsilon$ *for* $i = 1, \ldots, k$, *where* $\mathbf{x}_{si} = W\mathbf{y}_i$, *and* $\mathbf{x}_{di}$ *is a solution of* (4.2) *for* $i = 1, \ldots, k$.

Step 4. *Apply Step* 2 *to all single eigenvalues of* A_s *and Step* 3 *to all multiple eigenvalues of* A_s.

Many numerical methods for computing eigenvalues of Hermitian matrices in the original ring have been proposed. For real and complex Hermitian matrices, we may compute the eigenvalues by the power method, the symmetric QR algorithm, or the Jacobi method [18]. For quaternion Hermitian matrices, we may compute the eigenvalues by the power method [24, 28] or several state-of-the-art structure-preserving algorithms [23].

The linear systems (4.1) and (4.2) may be computed by several methods. Suppose the full eigenvalue decomposition of A_s is known as $A_s = U_s \Sigma_s U_s^*$. Then we have

$$\begin{aligned} \mathbf{x}_{di} &= (A_s - \lambda_s I)^+ (\lambda_d I - A_d)\mathbf{x}_{si} \\ &= U_s(\Sigma_s - \lambda_s I)^+ U_s^* (\lambda_d I - A_d)\mathbf{x}_{si} \\ &= -U_s(\Sigma_s - \lambda_s I)^+ U_s^* A_d \mathbf{x}_{si} \end{aligned} \tag{4.3}$$

for $i = 1, \ldots, k$. Otherwise, if the full eigenvalue decomposition of A_s is not known in advance, then we can solve the linear systems (4.1) and (4.2) directly. Both systems are consistent and solvable. It should be noted that a quaternion linear system may not be easy to solve since the quaternion numbers are not communicative. Researchers may reformulate the quaternion linear system as a real or complex linear system, which is four times or twice of the original dimension. They may also utilize the quaternion structure to improve the numerical performance. For instance, Li and Wang [27] proposed a structure-preserving quaternion biconjugate gradient method for solving non-Hermitian quaternion linear systems; Kyrchei [25] studied quaternion matrix equations by Cramer's rules; Fan et al. [16] proposed solving a quaternion linear system based on the semitensor product of quaternion matrices. Pan and Ng [30] studied a block-diagonalized formulation and proposed a fast algorithm for computing the inverse of a quaternion circulant matrix.

Furthermore, both systems (4.1) and (4.2) are ill-conditioned since the coefficient matrices are singular, which may lead to unstable numerical performance. In practice, note that for any $\alpha \in \mathbb{G}$ and $\mathbf{x}_d \in \mathbb{G}^n$ that satisfy (4.1), $\mathbf{x}_d(\alpha) = \mathbf{x}_d + \mathbf{x}_s\alpha$ is also a solution of (4.1). Thus, we may add an extra constraint $\mathbf{x}_s^*\mathbf{x}_d = 0$ and solve

$$\begin{bmatrix} A_s - \lambda_s I \\ \mathbf{x}_s^* \end{bmatrix} \mathbf{x}_d = \begin{bmatrix} (\lambda_d I - A_d)\mathbf{x}_s \\ 0 \end{bmatrix}. \tag{4.4}$$

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We call (4.4) the *augmented system* of (4.1). We may show the coefficient matrix of (4.4) has full column rank, and the augmented system admits one unique solution. We may also define the *augmented system* of (4.2) as

$$\begin{bmatrix} A_s - \lambda_s I \\ W^* \end{bmatrix} \mathbf{x}_{di} = \begin{bmatrix} (\lambda_d I - A_d)\mathbf{x}_{si} \\ O \end{bmatrix} \tag{4.5}$$

for $i = 1, \ldots, k$ to overcome the ill-condition issue. Here, $W = (\mathbf{v}_1 \ldots, \mathbf{v}_k)$.

Similarly, Algorithm 4.2 can be extended to compute several extreme eigenpairs or a few eigenpairs of a dual Hermitian matrix. One difficulty here is that if we do not know the multiplicity of the eigenvalue in advance, we may not obtain the eigenvectors exactly. One way to solve this is to compute several extra eigenvalues in the standard part.

5. The relative configuration problem in formation control. Consider the formation control problem of n rigid bodies. These n rigid bodies can be autonomous mobile robots, or unmanned aerial vehicles, or autonomous underwater vehicles, or small satellites. Then these n rigid bodies can be described by a graph $G = (V, E)$ with n vertices and m edges. For two rigid bodies i and j in V, if rigid body i can sense rigid body j, then edge $(i, j) \in E$. As studied in [38], we assume that any pair of these rigid bodies are mutual visual, i.e., rigid body i can sense rigid body j if and only if rigid body j can sense rigid body i. Furthermore, we assume that G is connected in the sense that for any node pair i and j, either $(i, j) \in E$ or there is a path connecting i and j in G, i.e., there are nodes $i_i, \ldots, i_k \in V$ such that $(i, i_1), \ldots, (i_k, j) \in E$.

Suppose that for each $(i, j) \in E$, we have a desired relative configuration from rigid body i to j as $q_{d_{ij}} \in \hat{\mathbb{U}}$. We say that the desired relative configurations $\left\{q_{d_{ij}} : (i, j) \in E\right\}$ is *reasonable* if and only if there is a desired formation $\mathbf{q}_d \in \hat{\mathbb{U}}^{n\times 1}$ which satisfies

$$q_{d_{ij}} = q_{d_i}^* q_{d_j} \tag{5.1}$$

for all $(i, j) \in E$.

Thus, a meaningful application problem in formation control is to verify whether a given desired relative configuration scheme is reasonable or not. See [29, 38, 44] for more study on formation control.

THEOREM 5.1. *Suppose that we have an undirected graph $G = (V, E)$, which is bidirectional and connected. The desired relative configuration $\left\{q_{d_{ij}} : (i, j) \in E\right\}$ is* reasonable *if and only if*

- *for all $(i, j) \in E$, we have*

$$q_{d_{ji}} = q_{d_{ij}}^*; \tag{5.2}$$

- *for any cycle $\{j_1, \ldots, j_k\}$, with $j_{k+1} = j_1$, of G, we have*

$$\prod_{i=1}^{k} q_{d_{j_i j_{i+1}}} = 1. \tag{5.3}$$

Furthermore, let $\mathbf{q}_0$ be a desired formation satisfying (5.1). *Then the set of all the desired formations satisfying* (5.1) *is*

$$\left\{c\mathbf{q}_0 : c \in \hat{\mathbb{U}}\right\}. \tag{5.4}$$

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Proof. Suppose that the desired relative configuration $\{q_{d_{ij}} : (i,j) \in E\}$ is reasonable. Then there is a desired formation $\mathbf{q}_d \in \hat{\mathbb{U}}^{n\times 1}$, which satisfies (5.1) for all $(i,j) \in E$. Then for all $(i,j) \in E$, we have $q_{d_{ji}} = q_{d_j}^* q_{d_i} = q_{d_{ij}}^*$, which implies (5.2). On the cycle $\{j_1, \ldots, j_k\}$, we have $q_{d_{j_{i+1}}} = q_{d_{j_i}} q_{d_{j_i j_{i+1}}}$ for $i = 1, \ldots, k$. This implies (5.3).

On the other hand, suppose that the desired relative configurations $\{q_{d_{ij}}\}$ satisfy (5.2) for all $(i,j) \in E$ and (5.3) for all cycles in G. We wish to show that there is a desired formation $\mathbf{q}_d \in \hat{\mathbb{U}}^{n\times 1}$, which satisfies (5.1) for all $(i,j) \in E$. We count two edges (i,j) and (j,i) as one bidirectional edge pair. Thus, G has $m_0 = \frac{m}{2}$ bidirectional edge pairs. By graph theory, since G is bidirectional and connected, we have $m_0 \geq n-1$. Let $M = m_0 - n + 1$. We now show by induction on M that there is a desired formation $\mathbf{q}_d \in \hat{\mathbb{U}}^{n\times 1}$, which satisfies (5.1) for all $(i,j) \in E$.

(i) We first show this for $M = 0$, i.e., $m_0 = n-1$. Then G is a bidirectional tree. Without loss of generality, we pick node 1 as the root of this tree. Let $q_{d_1} = 1$. Each node in $\{2, \ldots, n\}$ has one unique father node. Denote $V_{(1)}$ as the set of children nodes of the root. Let $q_{d_i} = q_{d_1} q_{d_{1i}} = q_{d_{1i}}$ for all $i \in V_{(1)}$. Similarly, denote $V_{(j)}$ as the set of children nodes of $V_{(j-1)}$. Then $q_{d_i} = q_{d_{f_i}} q_{d_{f_i i}}$ for all $i \in V_{(j)}$, where f_i is the father node of node i. We repeat this process until $V = \cup_j V_{(j)}$. By (5.2), (5.1) is satisfied for all $(i,j) \in E$.

(ii) We now assume that this claim is true for $M = M_0$ and prove that it is true for $M = M_0 + 1$. Then $M > m_0 - n + 1$. This implies that there is at least one cycle $\{j_1, \ldots, j_k\}$ with $j_{k+1} = j_1$, of G, such that the k nodes $j_1, \ldots, j_k$ are all distinct and $k \geq 3$. Delete the bidirectional edge pair (j_1, j_k) and (j_k, j_1). We get an undirected graph $G_0 = (V, E_0)$. Then G_0 is still bidirectional and connected, and the desired relative configurations $\{q_{d_{ij}} : (i,j) \in E_0\}$ satisfy (5.2) for all $(i,j) \in E_0$ and (5.3) for all cycles in G_0. By our induction assumption, there is a desired formation $\mathbf{q}_d \in \hat{\mathbb{U}}^{n\times 1}$ such that (5.1) holds for all $(i,j) \in E_0$. Now, by (5.2) for $i = j_1$ and $j = j_k$, and (5.3) for this cycle $\{j_1, \ldots, j_k\}$, (5.1) also holds for (j_1, j_k) and (j_k, j_1). This proves this claim for $M = M_0 + 1$ in this case.

Suppose that $\mathbf{q}_0$ is a desired formation satisfying (5.1), and $c \in \hat{\mathbb{U}}$. Then $c\mathbf{q}_0$ also satisfies (5.1). We may prove the other side of the last claim by induction as above. The proof is completed. ∎

The first condition in (5.2) is easy to verify. However, the second condition in (5.3) is relatively complicated since the number of cycles may increase exponentially with the number of nodes. Fortunately, we can solve this problem with the help of eigenvalues of dual Hermitian matrices, as shown in the next section.

Although we focus on the quaternion ring in this section, the results could also be generalized to the complex and real fields.

Before proceeding to the next section, we would like to highlight that eigenvalues of graph Laplacian have a long history in control law design and analysis. In 2004, Olfati-Saber and Murray [33] considered the linear system

$$\dot{\mathbf{x}}(t) = -L\mathbf{x}(t), \tag{5.5}$$

where L is the graph Laplacian induced by the weighted digraph $G = (V, E, A)$ and A is the weighted adjacency matrix. They showed that if G is a connected undirected graph or a strongly connected digraph, then (5.5) asymptotically solves a consensus problem from all initial states. The control law in (5.5) is simple yet powerful and has garnered significant attention. In 2020, Savino et al. [40] proposed a control rule based on dual quaternion to pose consensus of multiple rigid bodies. Recently, Qi and

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Cui [35] generalized the control law of pose consensus problem in [40] to a general pose formation control problem with arbitrary target configurations. The key to the asymptotic convergence of the control law is that zero is a simple eigenvalue of the Laplacian matrix L with a structured eigenvector, and all the other eigenvalues are on the open right half plane.

6. A cross-disciplinary approach for the relative configuration problem. In this section, we show that condition (5.3) is equivalent to the balance of the corresponding gain graph Φ [11]. Here, $\Phi = (G, \hat{\mathbb{G}}, \varphi)$ is a dual unit gain graph with n vertices, $G = (V, E)$ is the underlying graph, $n = |V|$, $m = |E|$, $\hat{\mathbb{G}}$ is the gain group, and $\varphi : E(\Phi) \to \hat{\mathbb{G}}$ is the gain function such that $\varphi(e_{ij}) = \varphi^{-1}(e_{ji})$. For instance, if $\hat{\mathbb{G}}$ is the set of unit dual complex numbers, then Φ is a dual complex unit gain graph (DCUGG). If $\hat{\mathbb{G}}$ is the set of unit dual quaternion numbers, then Φ is a dual quaternion unit gain graph (DQUGG). If $\hat{\mathbb{G}}$ is the set of unit real numbers, then Φ is a signed graph. The adjacency and Laplacian matrices of Φ are defined via the gain function $\varphi(e)$ as follows [11]:

$$a_{ij}(\Phi) = \begin{cases} \varphi(e_{ij}) & \text{if } e_{ij} \in E(\Phi), \\ 0 & \text{otherwise,} \end{cases} \quad \text{and } L(\Phi) = D - A(\Phi). \tag{6.1}$$

Here, $\varphi(e_{ij}) \in \hat{\mathbb{G}}$, $\varphi(e_{ij}) = \varphi(e_{ji})^{-1}$, and D the degree matrix of the corresponding underlying graph G. If $|\varphi(e_{ij})| = 1$ for all $e_{ij} \in E(\Phi)$, then $A(\Phi)$ and $L(\Phi)$ are Hermitian matrices in $\hat{\mathbb{G}}^{n\times n}$.

Given a dual unit gain graph $\Phi = (G, \hat{\mathbb{G}}, \varphi)$. A cycle is *neutral* if the product of all gains on this cycle is equal to one. An edge set is *balanced* if every cycle in this edge set is neutral. A graph is balanced if its edge set is balanced. A signed graph is in balance if all its cycles are positive, which was first studied by Frank Harary [21] and was applied in the study of social balance in social psychology. The balance of signed graphs was further studied in [46] and extended to complex unit gain graphs in [39], quaternion unit gain graphs in [3], and dual unit gain graphs in [11], respectively.

A *switching function* is a function $\zeta : V \to \hat{\mathbb{G}}$ that switches the $\hat{\mathbb{G}}$-gain graph $\Phi = (G, \hat{\mathbb{G}}, \varphi)$ to $\Phi^\zeta = (G, \hat{\mathbb{G}}, \varphi^\zeta)$, where

$$\varphi^\zeta(e_{ij}) = \zeta^{-1}(v_i)\varphi(e_{ij})\zeta(v_j).$$

In this case, Φ and Φ^ζ are switching equivalent, denoted by $\Phi \sim \Phi^\zeta$. A *potential function* is a function $\theta : V \to \hat{\mathbb{G}}$ such that for every $e_{ij} \in E(\Phi)$,

$$\varphi(e_{ij}) = \theta(v_i)^{-1}\theta(v_j). \tag{6.2}$$

We write $(G, 1_{\hat{\mathbb{G}}})$ as the $\hat{\mathbb{G}}$-gain graph with all 1-weighted edges.

The following result was obtained for signed graphs in [46], quaternion unit gain graphs in [3], and dual unit gain graphs in [11], respectively.

LEMMA 6.1. *Let $\Phi = (G, \hat{\mathbb{G}}, \varphi)$ be a gain graph. Then the following are equivalent:*
(i) *Φ is balanced.*
(ii) *$\Phi \sim (G, 1_{\hat{\mathbb{G}}})$.*
(iii) *φ has a potential function.*

We define $\varphi(e_{ij}) = q_{d_{ij}}$ for the formation control problem here. Then we may apply spectral graph theory. Let A and L be the adjacency and Laplacian matrices of Φ defined by (6.1), respectively. Recently, [11] showed that if Φ is balanced, then A and L are similar with the adjacency and Laplacian matrices of the underlying graph

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G, respectively, and its spectrum $\sigma_A(\Phi)$ consists of n real numbers, while its Laplacian spectrum $\sigma_L(\Phi)$ consists of one zero and $n-1$ positive numbers. Furthermore, there is $\sigma_A(\Phi)=\sigma_A(G)$ and $\sigma_L(\Phi)=\sigma_L(G)$.

Based on the above facts, we may verify the reasonableness of a desired relative configuration by computing all eigenvalues of the adjacency or Laplacian matrices of Φ. If $\sigma_A(\Phi)=\sigma_A(G)$ or $\sigma_L(\Phi)=\sigma_L(G)$, then the gain graph is balanced and the desired relative configuration is reasonable. Otherwise, the desired relative configuration is not reasonable. The method is applicable for low- and medium-dimensional problems.

In the following, we propose verifying the reasonableness of the desired relative configuration by computing the smallest eigenvalue of the Laplacian matrix.

THEOREM 6.2. *Let $\Phi=(G,\hat{\mathbb{G}},\varphi)$ be a gain graph with $n=|V|$, $m=|E|$. Suppose that $\varphi(e_{ij})=1$ for all $(ij)\in E$ and G has t subgraphs $G_i=(V_i,E_i)$, $n=n_1+\cdots+n_t$, $V_i=\{m_{i-1}+1,\ldots,m_i\}$, $m_i=n_1+\cdots+n_i$, and $n_0=0$. Let L be the Laplacian matrix of Φ defined by* (6.1)*. Then Φ is balanced if and only if the following conditions hold simultaneously:*

(i) *L has t zero eigenvalues that are the smallest eigenvalue of L.*
(ii) *For $i=1,\ldots,t$, their eigenvectors $\mathbf{x}_i\in\hat{\mathbb{G}}^n$ satisfies $|x_i(m_{i-1}+1)|=\cdots=|x_i(m_i)|$ and $x_i(j)=0$ for $j\notin[m_{i-1}+1,m_i]$.*
(iii) *Y^*LY is equal to the Laplacian matrix of the underlying graph G, where $Y=diag(\mathbf{y})\in\hat{\mathbb{G}}^{n\times n}$, $\mathbf{x}_a=\mathbf{x}_1+\cdots+\mathbf{x}_t$, and $y(i)=\frac{x_a(i)}{|x_a(i)|}$.*

Proof. The results follow directly from the fact that Φ is balanced if and only if there exists a diagonal matrix Y such that Y^*LY is equal to the Laplacian matrix of the underlying graph G and the spectral theory of the Laplacian matrix of a graph G. ∎

COROLLARY 6.3. *Let $\Phi=(G,\hat{\mathbb{G}},\varphi)$ be a gain graph with $n=|V|$, $m=|E|$. Suppose that $\varphi(e_{ij})=1$ for all $(ij)\in E$ and G is connected. Let L be the Laplacian matrix of Φ defined by* (6.1)*. Then Φ is balanced if and only if the following conditions hold simultaneously:*

(i) *L has a simple zero eigenvalue that is also the smallest eigenvalue of L.*
(ii) *The eigenvector $\mathbf{x}=(x(1),\ldots,x(n))^\top$ corresponding to the zero eigenvalue satisfies $|x(1)|=\cdots=|x(n)|$.*
(iii) *Y^*LY is equal to the Laplacian matrix of the underlying graph G, where $Y=diag(\mathbf{y})\in\hat{\mathbb{G}}^{n\times n}$ and $y(j)=\frac{x(j)}{|x(j)|}$ for all $j=1,\ldots,n$.*

Condition (i) is insufficient for Φ to be balanced, as demonstrated later in Example 7.3. This differs from the results for signed graphs and quaternion gain graphs. For instance, in 2003, Hou et al. [22] showed that a connected signed graph is balanced if and only if zero is an eigenvalue of the Laplacian matrix. A connected complex or quaternion gain graph is balanced if and only if zero is the smallest eigenvalue of the Laplacian matrix, as shown in [39] and [3], respectively.

A gain graph assigns an element of a mathematical group to each of its edges, and if a group element is assigned to an edge, then the inverse of that group element is always assigned to the inverse edge of that edge. If such a mathematical group consists of unit numbers of a number system, then the gain graph is called a unit gain graph. The real unit gain graph is called a signed graph [22]. There are studies on DCUGGs and DQUGGs [11]. The gain graph theory is a rich branch of algebraic graph theory. For instance, Hou, Li, and Pan [22] studied the largest Laplacian eigenvalue of a signed graph and Belardo et al. [3] studied the spectra of quaternion unit gain graphs. What

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we used here for the formation control problem are DQUGGs, which are not in the literature until the recent work by Cui et al. [11]. By exploring the eigenvalue problem of dual Hermitian matrices, and its link with the unit gain graph theory, we opened a cross-disciplinary approach to solve the relative configuration problem in formation control.

7. Numerical experiments. We begin with a toy example of computing all eigenpairs of the adjacency matrix of a DCUGG.

Example 7.1. Consider a dual complex unit gain cycle Φ_A in Figure 1(a). The adjacency matrix is given as follows:

$$A = \begin{bmatrix} 0 & 1+\mathbf{i}\epsilon & 1-2\mathbf{i}\epsilon \\ 1-\mathbf{i}\epsilon & 0 & 1-\mathbf{i}\epsilon \\ 1+2\mathbf{i}\epsilon & 1+\mathbf{i}\epsilon & 0 \end{bmatrix}.$$

In this example, the eigenvalues of A_s are $\lambda_{s1} = 2$, $\lambda_{s2} = \lambda_{s3} = -1$, and their corresponding eigenvectors are

$$\mathbf{v}_1 = \frac{\sqrt{3}}{3}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \frac{\sqrt{6}}{6}\begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \quad \mathbf{v}_3 = \frac{\sqrt{2}}{2}\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}. \tag{7.1}$$

The first eigenvalue is single. Thus $\mathbf{x}_{s1} = \mathbf{v}_1$, and

$$\lambda_{d1} = \mathbf{x}_{s1}^* A_d \mathbf{x}_{s1} = 0, \quad \mathbf{x}_{d1} = \frac{\sqrt{3}\mathbf{i}}{9}\begin{bmatrix} -1 \\ -2 \\ 3 \end{bmatrix}.$$

The standard parts of the second and the third eigenvalues are the same. Let $W = [\mathbf{v}_2, \mathbf{v}_3]$. Then the supplement matrix is

$$W^* A_d W = \frac{2\sqrt{3}}{3}\begin{bmatrix} 0 & \mathbf{i} \\ -\mathbf{i} & 0 \end{bmatrix},$$

and its eigenparis are

$$\lambda_{d2} = \frac{2\sqrt{3}}{3}, \quad \lambda_{d3} = -\frac{2\sqrt{3}}{3}, \quad \mathbf{y}_1 = \frac{\sqrt{2}}{2}\left[-1\mathbf{i}\right], \quad \mathbf{y}_2 = \frac{\sqrt{2}}{2}\left[1\mathbf{i}\right],$$

(a) Dual complex unit gain graph Φ_A. (b) Dual complex unit gain graph Φ_B.

FIG. 1. *Two DCUGGs.*

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respectively. Furthermore, there is

$$\mathbf{x}_{s2} = W\mathbf{y}_1 = \frac{\sqrt{3}}{6}\begin{bmatrix} -1+\sqrt{3}\mathbf{i} \\ -1-\sqrt{3}\mathbf{i} \\ 2 \end{bmatrix}, \quad \mathbf{x}_{s3} = W\mathbf{y}_2 = \frac{\sqrt{3}}{6}\begin{bmatrix} 1+\sqrt{3}\mathbf{i} \\ 1-\sqrt{3}\mathbf{i} \\ -2 \end{bmatrix}.$$

Last, we solve (4.2) and derive that

$$\mathbf{x}_{d2} = \frac{-1+3\sqrt{3}\mathbf{i}}{18}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{x}_{d3} = \frac{-1-3\sqrt{3}\mathbf{i}}{18}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Following the same approach, we derive the eigenpairs of the adjacency matrix of the dual complex unit gain cycle in Figure 1(b).

Example 7.2. Consider a dual complex unit gain cycle in Figure 1(b). The adjacency matrix is given as follows:

$$B = \begin{bmatrix} 0 & 1+\mathbf{i}\epsilon & 1 \\ 1-\mathbf{i}\epsilon & 0 & 1-\mathbf{i}\epsilon \\ 1 & 1+\mathbf{i}\epsilon & 0 \end{bmatrix}.$$

Its eigenvalues are

$$\lambda_1 = 2, \quad \lambda_2 = -1, \quad \lambda_3 = -1,$$

and their corresponding eigenvectors are

$$\mathbf{x}_1 = \frac{\sqrt{3}}{3}\begin{bmatrix} 1+\mathbf{i}/3\epsilon \\ 1-2\mathbf{i}/3\epsilon \\ 1+\mathbf{i}/3\epsilon \end{bmatrix}, \ \mathbf{x}_2 = \frac{\sqrt{6}}{6}\begin{bmatrix} 1-\mathbf{i}/3\epsilon \\ 1-\mathbf{i}/3\epsilon \\ -2-\mathbf{i}/3\epsilon \end{bmatrix}, \ \mathbf{x}_3 = \frac{\sqrt{2}}{2}\begin{bmatrix} 1+\mathbf{i}/3\epsilon \\ -1+\mathbf{i}/3\epsilon \\ \mathbf{i}/3\epsilon \end{bmatrix},$$

respectively.

Since the coefficient matrices in (4.1) and (4.2) are singular, the dual parts of the eigenvectors of B are not unique. For instance,

$$\tilde{\mathbf{x}}_1 = \frac{\sqrt{3}}{3}\begin{bmatrix} 1 \\ 1-\mathbf{i}\epsilon \\ 1 \end{bmatrix}, \ \tilde{\mathbf{x}}_2 = \frac{\sqrt{6}}{6}\begin{bmatrix} 1 \\ 1-\mathbf{i}\epsilon \\ -2 \end{bmatrix}, \ \tilde{\mathbf{x}}_3 = \frac{\sqrt{2}}{2}\begin{bmatrix} 1 \\ -1+\mathbf{i}\epsilon \\ 0 \end{bmatrix}$$

are also eigenvectors of B and we have $\tilde{\mathbf{x}}_1 = \mathbf{x}_1 - \frac{\mathbf{i}}{3}\mathbf{x}_{s1}\epsilon$, $\tilde{\mathbf{x}}_2 = \mathbf{x}_2 - \frac{\mathbf{i}}{6}\mathbf{x}_{s2}\epsilon + \frac{\sqrt{3}\mathbf{i}}{6}\mathbf{x}_{s3}\epsilon$, and $\tilde{\mathbf{x}}_2 = \mathbf{x}_2 + \frac{\sqrt{3}\mathbf{i}}{6}\mathbf{x}_{s2}\epsilon - \frac{\mathbf{i}}{2}\mathbf{x}_{s3}\epsilon$.

We note that all eigenvalues of B are real numbers. This is because Φ_B is balanced [11]. In fact, let $B^{(0)}$ be the adjacency matrix of the underlying graph G. Then the eigenvalues of B are equal to those of $B^{(0)}$. Furthermore, let $P = \mathrm{diag}(1, 1-\mathbf{i}\epsilon, 1)$. Then we have

$$B = PB^{(0)}P^*$$

and $\tilde{\mathbf{x}}_1 = P\mathbf{v}_1$, $\tilde{\mathbf{x}}_2 = P\mathbf{v}_2$, $\tilde{\mathbf{x}}_3 = P\mathbf{v}_3$, where $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$ are defined in (7.1). This is corresponding to the theory that if $B^{(0)}\mathbf{v}_i = \lambda_i\mathbf{v}_i$, then $BP\mathbf{v}_i = \lambda_i P\mathbf{v}_i$ for $i = 1, \ldots, 3$.

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7.1. Eigenvalues of balanced dual unit gain cycles. We continue to test large-scale dual complex and dual quaternion unit gain cycles. We use the default command "eig" in MATLAB to compute all eigenvalues and eigenvectors of a complex matrix and the package "qtmf"[1] to compute all eigenpairs of a quaternion matrix. We first generate n unit dual complex or dual quaternion numbers q_i, $i = 1, \ldots, n$. Then we define the gain of each edge to be $\phi(e_{i,i+1}) = q_i^* q_{i+1}$ for $i = 1, \ldots, n-1$, and $\phi(e_{n,1}) = q_n^* q_1$. In this way, the corresponding cycles are balanced. For balanced dual complex unit gain cycles and dual quaternion unit gain cycles, the eigenvalues of the Laplacian matrices have closed from solutions [11] as follows:

$$\sigma_L = \left\{ 2 - 2\cos\left(\frac{2\pi j}{n}\right) : \ j \in \{0, \ldots, n-1\} \right\}. \tag{7.2}$$

Let the number of vertices be $n \in \{10, 20, 50, 100, 200, 500\}$. We generate random unit dual elements as the gains and then compute the eigenpairs by Algorithm 4.2. The subproblems (4.1) and (4.2) in Algorithm 4.2 are solved by two approaches: the generalized inverse solution in (4.3) and the augmented systems in (4.4) and (4.5). Both real and complex augmented systems are solved directly by the MATLAB command "linsolve." For the quaternion augmented system, we first reformulate the quaternion linear system into a real system (see [12] for more details) and solve the resulting system by the MATLAB command "linsolve."

We also compare our Algorithm 4.2 with the power method proposed in [12]. When computing all eigenpairs, we first implement the power method on L to compute the first eigenpairs $(\lambda_1, \mathbf{x}_1)$ and then implement the power method on $L^{(1)} = L - \lambda_1 \mathbf{x}_1 \mathbf{x}_1^*$ to compute the second eigenpairs $(\lambda_2, \mathbf{x}_2)$. We repeat this process to compute all eigenpairs sequentially. We stop the power method when $\|L\mathbf{x}^k - \lambda^k \mathbf{x}^k\|_{2^R} \leq 10^{-10}\|L\|_{F^R}$ or the maximal iteration number 1000 is reached. We use "$-$" to denote the algorithm fails to return the result in 10^4 seconds. Define the computational residue by

$$\mathrm{RES} = \frac{1}{n}\|(\lambda)_{\mathrm{sorted}} - (\sigma_L)_{\mathrm{sorted}}\|_{2^R} + \frac{1}{n}\|LX - X\mathrm{diag}(\lambda)\|_{F^R},$$

where $\lambda = [\lambda_1, \ldots, \lambda_n]^\top$ and $X = [\mathbf{x}_1, \ldots, \mathbf{x}_n]$ are the eigenvalues and eigenvectors returned by the algorithms, and $(\lambda)_{\mathrm{sorted}}$ and $(\sigma_L)_{\mathrm{sorted}}$ are the sorted eigenvalues from numerical experiments and (7.2) in the descending order, respectively.

In Table 1, we report the CPU time in seconds and the residue for computing all eigenvalues of the Laplacian matrices of DCUGGs and DQUGGs. From this table, we can see all methods can return accurate results in 6 s when $n = 10$ or 20. However, when n is larger than 50, our proposed supplement matrix method method is faster and more accurate than the power method. The larger the scale, the more obvious the trend. Both (4.3) and (4.5) are efficient and accurate for solving the linear systems in the dual part. In comparison, (4.5) is preferable for the complex case and (4.3) is better for the quaternion case. When $n = 500$, the power method fails to return the result in 10^4 seconds. The possible reason is that the convergence rate of the power method is proportional to the absolute value of the standard part of the second largest eigenvalue divided by the largest eigenvalue, which shall converge to 1 when n goes to infinity. Consequently, the convergence rate of the power method will be slow as n increases.

[1] https://qtfm.sourceforge.io/.

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TABLE 1
Numerical results for computing all eigenpairs of dual complex and dual quaternion unit gain cycles.

Method	n	10	20	50	100	200	500
				DCUGG			
Alg. 4.2	CPU	7.36e−04	1.36e−03	8.00e−03	5.05e−02	6.66e−01	2.00e+01
+(4.3)	RES	1.02e−15	1.23e−15	2.10e−15	3.10e−15	5.36e−15	1.06e−14
Alg. 4.2	CPU	4.43e−04	1.63e−03	6.71e−03	4.94e−02	7.51e−01	1.26e+01
+(4.5)	RES	9.75e−16	9.09e−16	7.86e−16	4.87e−16	3.10e−16	2.41e−16
Power method	CPU	2.19e−02	1.11e−01	7.32e−01	4.39e+00	3.09e+01	3.75e+02
	RES	2.91e−10	4.84e−10	2.98e−05	2.66e−04	2.84e−04	2.33e−03
				DQUGG			
Alg. 4.2	CPU	3.03e−02	1.43e−01	1.61e−01	8.32e−01	3.47e+00	1.17e+02
+(4.3)	RES	1.31e−15	1.19e−15	1.35e−15	1.80e−15	4.40e−15	5.87e−15
Alg. 4.2	CPU	4.37e−02	1.19e−01	2.74e−01	1.35e+00	9.47e+00	4.38e+02
+(4.5)	RES	1.66e−15	1.39e−15	2.12e−15	2.04e−15	3.29e−15	3.20e−15
Power method	CPU	2.23e+00	5.65e+00	3.95e+01	1.91e+02	2.06e+03	–
	RES	2.29e−10	1.42e−10	9.42e−05	7.95e−05	7.99e−05	–

7.2. Balance of dual unit gain graphs. We continue to verify the balance of the dual unit gain graphs by Theorem 6.2. We check the first two conditions of Theorem 6.2 manually and define the residue of the third condition by

$$Err = \|Y^* L Y - L^{(0)}\|_{F^R}. \tag{7.3}$$

Here, L and $L^{(0)}$ are the Laplacian matrices of the gain graph and the underlying graph, respectively. We say the gain graph is balanced if the first two conditions of Theorem 6.2 hold true and Err is less than a threshold, i.e., 10^{-8} in our numerical experiments.

Example 7.3. We first verify the balance of Φ_A and Φ_B in Examples 7.1 and 7.2. The Laplacian matrices of G, Φ_A, Φ_B are

$$L^{(0)} = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}, \quad L_A = \begin{bmatrix} 2 & -1-\mathbf{i}\epsilon & -1+2\mathbf{i}\epsilon \\ -1+\mathbf{i}\epsilon & 2 & -1+\mathbf{i}\epsilon \\ -1-2\mathbf{i}\epsilon & -1-\mathbf{i}\epsilon & 2 \end{bmatrix},$$

and

$$L_B = \begin{bmatrix} 2 & -1-\mathbf{i}\epsilon & -1 \\ -1+\mathbf{i}\epsilon & 2 & -1+\mathbf{i}\epsilon \\ -1 & -1-\mathbf{i}\epsilon & 2 \end{bmatrix},$$

respectively. The eigenvalues of L_A are $\{0, 3+1.1547\epsilon, 3-1.1547\epsilon\}$ and the eigenvector corresponding to the zero eigenvalue is

$$\mathbf{x}_A = \frac{\sqrt{3}}{3}\left[1 - \frac{\mathbf{i}}{3}\epsilon, \quad 1 - \frac{2\mathbf{i}}{3}\epsilon, \quad 1 + \mathbf{i}\epsilon\right]^\top.$$

From this, we conclude that conditions (i) and (ii) in Theorem 6.2 hold true. For condition (iii), it follows from (7.3) that $Err_A = 1.6330$. Hence, Φ_A is not balanced.

The eigenvalues of L_B are $\{0, 3, 3\}$, and the eigenvector corresponding to the zero eigenvalue is

$$\mathbf{x}_B = \frac{\sqrt{3}}{3}\left[1 + \frac{\mathbf{i}}{3}\epsilon, \quad 1 - \frac{2\mathbf{i}}{3}\epsilon, \quad 1 + \frac{\mathbf{i}}{3}\epsilon\right]^\top.$$

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TABLE 2
Numerical results for verifying the balance of the dual unit gain cycles.

n	10	20	50	100	200	500
			DCUGG			
CPU (s)	7.78e−04	1.79e−03	1.36e−02	8.15e−03	5.54e−02	1.96e−01
Err	6.92e−15	3.02e−14	4.04e−14	1.69e−13	1.34e−13	1.36e−12
			DQUGG			
CPU (s)	1.20e−02	2.60e−02	7.47e−02	2.57e−01	1.28e+00	4.13e+01
Err	9.81e−15	2.35e−14	4.96e−14	1.20e−13	4.50e−13	1.47e−12

From this, we see that conditions (i) and (ii) in Theorem 6.2 hold true and $Err_B = 2.22e-16$. Hence, Φ_B is balanced.

This example shows that zero is an eigenvalue of the Laplacian matrix is not sufficient for the dual unit gain graph to be balanced, which further demonstrates the difference between the balance theory of dual unit gain graph and its unit gain graph counterpart.

We continue to test the cycles in subsection 7.1. All these cycles are balanced and the numerical results are shown in Table 2. From this table, we see that the residue is less than 10^{-11} for all examples, and all results can be obtained in 5 seconds. Therefore, we can confirm these cycles are balanced.

Furthermore, we test unbalanced cycles with $n \in \{10, 20, 50, 100, 200, 500\}$. We conduct experiments on two groups of cycles: (a) let $\phi(e_{ij})$ be a random unit dual complex or quaternion number if $i - j = 1$ or $i - j = 1 - n$ and $\phi(e_{ji}) = \phi(e_{ij})^*$ for $i, j = 1, \ldots, n$; (b) let the standard parts of $\phi(e_{ij})$ be one, the dual parts of $\phi(e_{ij})$ be imaginary complex or quaternion numbers, and $\phi(e_{ji}) = \phi(e_{ij})^*$ for $i, j = 1, \ldots, n$. In our numerical experiments, for all examples in the group (a), the smallest eigenvalues are not zero, hence violating condition (i) in Theorem 6.2. For all examples in the group (b), conditions (i) and (ii) in Theorem 6.2 hold, yet condition (iii) in Theorem 6.2 fails because the error defined in (7.3) is always greater than 0.05. In other words, all these cycles are unbalanced. These results show that we can verify the balance of dual unit gain graphs efficiently.

8. Concluding remarks. In this paper, we studied dual number symmetric matrices, dual complex Hermitian matrices, and dual quaternion Hermitian matrices in a unified frame as dual Hermitian matrices. This avoided unnecessary repetitions. We then proposed a practical method—the supplement matrix method—for computing eigenvalues of a dual Hermitian matrix. We raised a meaningful application problem—the relative configuration problem—in multiagent formation control. A cross-disciplinary approach to solve the above problem is explored. This approach combines the spectral theory of dual Hermitian matrices and the unit gain graph theory. While the unit gain graph theory is well-developed in the spectral graph area, what we used here are about DQUGGs. Finally, the supplement matrix method was used in this approach.

REFERENCES

[1] J. ANGELES, *The dual generalized inverses and their applications in kinematic synthesis*, in Latest Advances in Robot Kinematics, Springer, Dordrecht, 2012, pp. 1–12.

[2] Z. J. BAI, D. CHU, AND D. F. SUN, *A dual optimization approach to inverse quadratic eigenvalue problems with partial eigenstructure*, SIAM J. Sci. Comput., 29 (2007), pp. 2531–2561.

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[3] F. Belardo, M. Brunetti, N. J. Coble, N. Reff, and H. Skogman, *Spectra of quaternion unit gain graphs*, Linear Algebra Appl., 632 (2022), pp. 15–49.

[4] G. Brambley and J. Kim, *Unit dual quaternion-based pose optimization for visual runway observations*, IET Cyber-Systems Robotics, 2 (2020), pp. 181–189.

[5] S. Bultmann, K. Li, and U. D. Hanebeck, *Stereo visual SLAM based on unscented dual quaternion filtering*, in Proceedings of the 22nd International Conference on Information Fusion, 2019, pp. 1–8.

[6] J. Cheng, J. Kim, Z. Jiang, and W. Che, *Dual quaternion-based graph SLAM*, Robot. Auton. Syst., 77 (2016), pp. 15–24.

[7] D. Chu and G. H. Golub, *On a generalized eigenvalue problem for nonsquare pencils*, SIAM J. Matrix Anal. Appl., 28 (2006), pp. 770–787.

[8] E. K.-W. Chu, *Perturbation of eigenvalues for matrix polynomials via the Bauer–Fike theorems*, SIAM J. Matrix Anal. Appl., 25 (2003), pp. 551–573.

[9] E. K.-W. Chu, H.-Y. Fan, and W.-W. Lin, *Projected generalized discrete-time periodic Lyapunov equations and balanced realization of periodic descriptor systems*, SIAM J. Matrix Anal. Appl., 29 (2007), pp. 982–1006.

[10] W. K. Clifford, *Preliminary sketch of bi-quaternions*, Proc. Lond. Math. Soc., 4 (1873), pp. 381–395.

[11] C. Cui, Y. Lu, L. Qi, and L. Wang, *Spectral properties of dual unit gain graphs*, Symmetry, 16 (2024), 1142.

[12] C. Cui and L. Qi, *A power method for computing the dominant eigenvalue of a dual quaternion Hermitian matrix*, J. Sci. Comput., 100 (2024).

[13] K. Daniilidis, *Hand-eye calibration using dual quaternions*, Int. J. Robot. Res., 18 (1999), pp. 286–298.

[14] W. Ding, Y. Li, T. Wang, and M. Wei, *Dual quaternion singular value decomposition based on bidiagonalization to a dual number matrix using dual quaternion Householder transformations*, Appl. Math. Lett., 152 (2024), 109021.

[15] A. Q. Duan, Q. W. Wang, and X. F. Duan, *On Rayleigh Quotient Iteration for Dual Quaternion Hermitian Eigenvalue Problem*, https://arxiv.org/abs/2310.20290, 2024.

[16] X. Fan, Y. Li, Z. Liu, and J. Zhao, *Solving quaternion linear system based on semi-tensor product of quaternion matrices*, Symmetry, 14 (2022), 1359.

[17] I. S. Fischer, *Dual-Number Methods in Kinematics, Statics and Dynamics*, CRC Press, Boca Raton, FL, 1998.

[18] G. H. Golub and C. F. Van Loan, *Matrix Computations*, 4th ed., Johns Hopkins University Press, Baltimore, MD, 2013.

[19] Y. L. Gu and L. Luh, *Dual-number transformation and its applications to robotics*, IEEE J. Robot. Automat., 3 (1987), pp. 615–623.

[20] I. Hadi, T. Bendory, and N. Sharon, *$SE(3)$ Synchronization by eigenvectors of dual quaternion matrices*, Inf. Inference, 13 (2024), iaae014.

[21] F. Harary, *On the notion of balanced in a signed graph*, Michigan Math. J., 2 (1953), pp. 143–146.

[22] Y. P. Hou, J. S. Li, and Y. L. Pan, *On the Laplacian eigenvalues of signed graphs*, Linear Multilinear Algebra, 51 (2003), pp. 21–30.

[23] Z. Jia, *The Eigenvalue Problem of Quaternion Matrix: Structure-Preserving Algorithms and Applications*, Science Press, Beijing, 2019.

[24] Z. Jia, Q. Wang, H. Pang, and M. Zhao, *Computing partial quaternion egenpairs with quaternion shift*, J. Sci. Comput., 97 (2023).

[25] I. Kyrchei, *Explicit representation formulas for the minimum norm least squares solutions of some quaternion matrix equations*, Linear Algebra Appl., 438 (2013), pp. 136–152.

[26] G. Leclercq, Ph Lefévre, and G. Blohm, *3D kinematics using dual quaternions: Theory and applications in neuroscience*, Front. Behav. Neurosci., 7 (2013), pp. 1–25.

[27] T. Li and Q.-W. Wang, *Structure preserving quaternion biconjugate gradient method*, SIAM J. Matrix Anal. Appl., 45 (2024), pp. 306–326.

[28] Y. Li, M. Wei, F. Zhang, and J. Zhao, *On the power method for quaternion right eigenvalue problem*, J. Comput. Appl. Math., 345 (2019), pp. 59–69.

[29] Z. Lin, L. Wang, Z. Han, and M. Fu, *Distributed formation control of multi-agent systems using complex Laplacian*, IEEE Trans. Automat. Control, 59 (2014), pp. 1765–1777.

[30] J. Pan and M. Ng, *Block-diagonalization of quaternion circulant matrices with applications*, SIAM J. Matrix Anal. Appl., 45 (2024), pp. 1429–1452.

[31] E. Pennestrì and R. Stefanelli, *Linear algebra and numerical algorithms using dual numbers*, Multibody Syst. Dyn., 18 (2007), pp. 323–344.

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[32] E. Pennestri and P. P. Valentini, *Linear dual algebra algorithms and their applications to kinematics*, in Multibody Dynamics, Springer, Dordrecht, 2009, pp. 207–229.

[33] R. Olfati-Saber and R. M. Murray, *Consensus problems in networks of agents with switching topology and time-delays*, IEEE Trans. Automat. Control, 49 (2004), pp. 1520–1533.

[34] L. Qi and C. Cui, *Eigenvalues and Jordan forms of dual complex matrices*, Commun. Appl. Math. Comput., 4 (2023), https://doi.org/10.1007/s42967-023-00299-1.

[35] L. Qi, C. Cui, and C. Ouyang, *Dual Quaternion Weighted Directed Graph and Formation Control*, arXiv:2401.05132v5, 2024.

[36] L. Qi, C. Ling, and H. Yan, *Dual quaternions and dual quaternion vectors*, Commun. Appl. Math. Comput., 4 (2022), pp. 1494–1508.

[37] L. Qi and Z. Luo, *Eigenvalues and singular values of dual quaternion matrices*, Pac. J. Optim., 19 (2023), pp. 257–272.

[38] L. Qi, X. Wang, and Z. Luo, *Dual quaternion matrices in multi-agent formation control*, Commun. Math. Sci., 21 (2023), pp. 1865–1874.

[39] N. Reff, *Spectral properties of complex unit gain graphs*, Linear Algebra Appl., 436 (2012), pp. 3165–3176.

[40] H. J. Savino, L. C. A. Pimenta, J. A. Shah, and B. V. Adorno, *Pose consensus based on dual quaternion algebra with application to decentralized formation control of mobile manipulators*, J. Franklin Inst., 357 (2020), pp. 142–178.

[41] E. Study, *Geometrie der Dynamen*, Verlag Teubner, Leipzig, 1903.

[42] F. E. Udwadia, *Dual generalized inverses and their use in solving systems of linear dual equations*, Mech. Mach. Theory, 156 (2021), 104158.

[43] H. Wang, *Characterization and properties of the MPDGI and DMPGI*, Mech. Mach. Theory, 158 (2021), 104212.

[44] X. Wang, C. Yu, and Z. Lin, *A dual quaternion solution to attitude and position control for rigid body coordination*, IEEE Trans. Robot., 28 (2012), pp. 1162–1170.

[45] T. Wei, W. Ding, and Y. Wei, *Singular value decomposition of dual matrices and its application to traveling wave identification in the brain*, SIAM J. Matrix Anal. Appl., 45 (2024), pp. 634–660.

[46] T. Zaslavsky, *Biased graphs.* I. *Bias, balance, and gains*, J. Combin. Theory Ser. B, 47 (1989), pp. 32–52.

[47] F. Zhang, *Quaternions and matrices of quaternions*, Linear Algebra Appl., 251 (1997), pp. 21–57.

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